

Project Proposal

1 Project Title

Vision-Based Terrain Classification for Robotic Navigation Using Classical Machine Learning

2 Team Members

- Aaron Miller

3 Project Details

3.1 Project Objective

- **Objective:**
To classify terrain types (e.g., grass, gravel, asphalt, dirt) from image patches using classical computer vision feature extraction and machine learning methods.
- **Problem:**
Autonomous robots and mobile systems must distinguish between traversable and non-traversable terrain to navigate safely. Terrain classification is a perception problem where the robot must infer surface properties from visual input.
- **Importance:**
Mobile autonomous systems are increasingly deployed in outdoor and unstructured environments where terrain conditions directly impact mobility and safety. The ability to automatically classify terrain types from visual input provides critical situational awareness, enabling safer navigation, improved path planning, and more reliable deployment in research, industrial, and civil settings.
- **Why machine learning helps:**
Terrain appearance varies significantly due to:
 - Lighting
 - Texture
 - Surface Roughness
 - Viewpoint

Hand-coded rules are brittle. Machine learning enables classification based on statistical texture and gradient features learned from data.

3.2 Datasets

I will use the RUGD (Robotics Urban Ground Dataset), a publicly available dataset collected from a ground mobile robot in outdoor environments. The dataset contains pixel-level annotations of terrain types such as asphalt, grass, dirt, and gravel. For this project, we will extract fixed-size 64×64 pixel patches from the raw RGB images and assign terrain labels based on the dominant annotated class within each patch, converting the segmentation dataset into a multiclass patch classification problem. This yields tens of thousands of labeled examples, which will be split approximately 70/15/15 into training, validation, and testing sets.

Each image patch serves as the input to a feature extraction pipeline rather than being fed directly to the classifier as raw pixels. Two classical texture descriptors will be computed per patch: Histogram of Oriented Gradients (HOG), which encodes the local distribution of edge orientations and captures structural texture, and Local Binary Patterns (LBP), which encodes micro-texture by comparing each pixel’s intensity to its neighbors and computing a histogram of the resulting binary codes. These two descriptors are concatenated to form a single high-dimensional feature vector that serves as the feature representation for each training example. The terrain class label is the corresponding target output. This approach captures texture statistics that are robust to moderate changes in lighting and viewpoint, rather than relying on raw pixel values that would not generalize across imaging conditions.

3.3 Machine Learning Algorithm

This project will use a Support Vector Machine (SVM) classifier trained on texture-based image features extracted from the RUGD dataset. Histogram of Oriented Gradients (HOG) and Local Binary Patterns (LBP) will be used to represent each image as a high-dimensional feature vector capturing edge structure and local texture information.

An SVM is chosen because it constructs a decision boundary that not only separates terrain classes but maximizes the margin between them, leading to stronger generalization on unseen data. Since HOG and LBP produce high-dimensional feature representations and the dataset size is moderate, SVMs provide an effective balance between model capacity and overfitting control.

Both a linear SVM and an RBF-kernel SVM will be evaluated. The linear model tests whether terrain categories are separable using a simple hyperplane in feature space, while the RBF kernel allows nonlinear decision boundaries if texture distributions overlap more complexly. Comparing these variants will help determine the intrinsic separability of terrain classes under classical texture descriptors.

3.4 Expected Outcomes

The expected outcome is a terrain classification model capable of distinguishing outdoor surface types (e.g., grass, gravel, dirt, asphalt) using texture-based features with strong generalization performance.

It is anticipated that:

- The RBF SVM will outperform the linear SVM if terrain classes exhibit nonlinear overlap.
- Texture-heavy classes (e.g., gravel vs. asphalt) will be more easily separable than visually similar classes (e.g., dirt vs. mud).

The project should demonstrate that classical computer vision features combined with margin-based classifiers can effectively recognize outdoor terrain types without requiring deep learning, providing a computationally efficient baseline for robotics-oriented perception tasks.